

is that the resistive interface between the electrolyte and the cathode commonly formed during the cycle can be reversed, thus extending the cycle life. The second is related to the flexibility of the organic cathode which allows it to maintain intimate contact at the interface with the solid electrolyte, even as the cathode expands and contracts during the cycle. This serves as a solution to one of the major problems till now.

The organic cathode is much more flexible, maintaining contact at all times with the interface and improving cycle life. According to the researchers, the contact between the electrode and electrolyte remained steady through at least 200 cycles, and this reliable contact will have a great chance of creating a high-performance solid-state battery.

Will sodium replace lithium?

Though Li-ion batteries play an important role in the transition towards a low-carbon economy, as the market rapidly expands, so does the need. The environmental and social challenges associated with the mass production of Li-ion batteries is triggering attention towards the search for alternative energy storage solutions based on materials that can be sourced in a sustainable and responsible way.

Na-ion batteries have great potential to represent the next-generation low-cost and environmentally friendly energy storage solution. The varied key performance indicators required by different applications and the market diversification are the driving forces pushing the Na-ion technology closer to the market. Na-ion batteries offer a combination of attractive properties as they are low-cost, use sustainable precursors, and have secure raw material supplies.

Na-ion batteries do not represent a massive departure from their Li-ion variants as the elemental structure of sodium is quite similar to lithium (being a Group 1 metal); thus material testing processes are similar and manufacturing processes are comparable too. Sodium is many magnitudes more abundant than lithium and is found everywhere in the Earth's crust and is harvestable from the ocean.

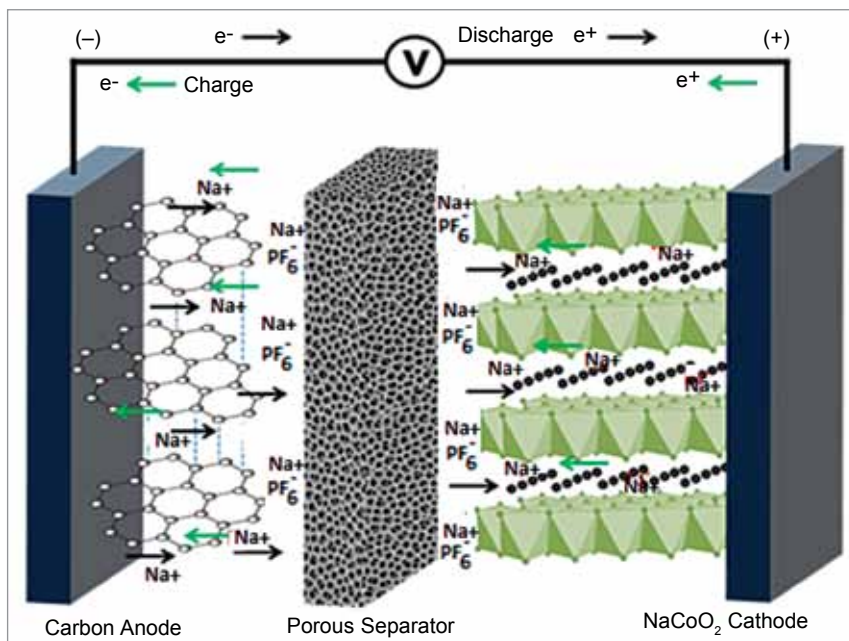


Fig. 2: Sodium-ion battery (Credit: ACS Energy Letters)

Thus, Na-ion batteries are considered as a drop-in technology that could benefit from the already existing Li-ion batteries manufacturing facilities.

Similar to Li-based systems, Na-based batteries also come in different forms such as Na-ion, Na-all-solid-state-battery, NaO₂ and Na/S. Na-ion technology represents an attractive technology almost ready to challenge the Li-ion batteries in specific applications, while the other ones are seen as disruptive future technologies.

A promising solution for stationary energy storage systems

According to researchers working for the development of batteries, with the ongoing developmental growth, the present best materials available for Na-ion cells should allow approaching the energy density of the present generation of Li-ion commercial cells. At present, sodium batteries are not quite viable for personal, portable electronics but are better suited for stationary applications.

One of the most significant application areas for the Na-ion battery prototype is certainly stationary energy storage systems, where cost and cycle-life represent two fundamental parameters. They have the potential to dominate the future market by representing as the most promising

system to fill the gap between energy production and utilisation by securing energy supply.

At lesser cost, consuming lesser resources

Ironically, the similarities between both chemistries and production strategies are a boon for Na-ion batteries. While cobalt isn't as integral to Na-ion batteries, sodium cobalt oxide (NaCoO₂) remains a common cathode material and researchers have been able to develop a new sodium oxide cathode material.

Additionally, these experimental sodium-ion batteries retain capacity, resist moisture, and have little voltage-fade, which is a common phenomenon with prolonged lithium battery cycles. Thus, Na-ion rechargeable batteries could soon be a cheaper and resource-saving alternative to current lithium-ion batteries.

As the research stands

At present, the electromobility version of transportation, especially in combination with renewable electricity utilisation, is regarded world-wide as a climate-friendly model of transportation for the future. However, the intense focus on the use of the rare and expensive lithium metal in rechargeable batteries has slowed down the large-scale electrification of